**1** **(a)** $15 - 3 \times 6$ or $15 - 18$ **[M1]**

$-3$ **[A1]**

**(b)** $\frac{9}{4} \times \frac{8}{3}$ **[M1]**

6 **[A1]**

$4\frac{1}{2}$ or $\frac{9}{2}$ **[A1]**

*[Total Marks: 5]*

**2** Denominator: $2 - 0.4 = 1.6$ or $\frac{8}{5}$ **[B1]**

Right hand side: $-1\frac{1}{4} = -\frac{5}{4}$ **[B1]**

$x + \frac{1}{3} = -\frac{5}{4} \times \frac{8}{5}$ **[M1]**

$x + \frac{1}{3} = -2$ **[A1]**

$x = -\frac{7}{3}$ or $-2\frac{1}{3}$ **[A1]**

*[Total Marks: 5]*

**3** Net rate of fill $= 0.75 - 0.45 = 0.3$ litres/min oe **[M1]**

Total time $= \frac{85.5}{0.3} = 285$ minutes **[M1]**

$285 \div 60 = 4$ with remainder 45 **[M1]**

4 hours 45 minutes **[A1]**

*[Total Marks: 4]*

**4** **(a)** 9 **[B1]**

**(b)** 25 **[B1]**

**(c)** 27 **[B2]**

If 0 scored, M1 for $-4$ or $-3$ seen **[M1]**

*[Total Marks: 4]*

**5** **(a)** $\frac{8}{6} - \frac{5}{6}$ or $\frac{1}{2}$ **[M1]**

$\frac{12}{5} \div$ their $\frac{1}{2}$ **[M1]**

$\frac{12}{5} \times \frac{2}{1}$ or equivalent multiplication **[M1]**

$4\frac{4}{5}$ **[A1]**

**(b)** $0.54$ or $\dfrac{45 \times 12}{600}$ or equivalent **[M1]**

$54 \div 6$ or $0.54 \div 0.06$ **[M1]**

9 **[A1]**

*[Total Marks: 7]*

**6 (a)** $\dfrac{2}{5} \times 1000$ or 400 **[M1]**

$18.5 \times 20$ or 370 **[M1]**

770 **[A1]**

**(b)** Target volume $= 500$ or volume to lose $= 770 - 500 = 270$ **[M1]**

$270 \div 4.5$ **[M1]**

60 **[A1]**

**(c)** Volume to fill $= 1000 - 500 = 500$ **[M1]**

1 hour 20 minutes $= 80$ minutes **[M1]**

$500 \div 80$ leading to 6.25 **[A1]**

*[Total Marks: 9]*